

Vikram Kommera

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EDUCATION

University of California, San Diego Sep 2025 - Jun 2028
Bachelors of Science in Electrical Engineering, minor in Mathematics GPA: 3.90/4.0

EXPERIENCE

Qualcomm Institute - Cultural Heritage Engineering Initiative Nov. 2025 - Present
Research Assistant San Diego, California

- Working on robotic systems to map and preserve coral reefs off the coast of Saudi Arabia
- Developing underwater VSLAM (Visual Simultaneous Localization and Mapping), developed a port of Stereo-Inertial ORB-SLAM3 for ROS2 Humble
- Created custom low-cost stereo-inertial sensor system using RasPi global shutter cameras and BNO055 IMU

Triton Robotics Aug. 2025 - Present
Autonomy Team San Diego, California

- Developed solutions for autonomous robot localization and navigation for the Advanced Robotics Competition (ARC)
- Implemented 4D LiDAR sensor with Point-LIO for accurate odometry with <1cm error at 4k+ Hz
- Developed a reliable, performant, end-to-end navigation stack using PID-to-point and Guided Vector Fields (GVF) completely in C++

Axiom Robotics (Founder's Inc 25) Jun. 2025 - Oct. 2025
Robotics Engineer San Francisco, California

- Created a custom built delta robot for automated quality control on industrial assembly lines
- Robot able to pick up over 80 defective objects per minute, with 96% detection accuracy
- Participated in Founders Inc. Off-Season program, a prestigious startup accelerator program in San Francisco

Dublin Robotics Club Dec. 2023 - May 2024
Robotics Coach, Part Time Dublin, California

- Worked with students to develop control solutions such as RAMSETE and Pure Pursuit.
- Built autonomous systems for VEX AI with path planning, obstacle avoidance, and object detection; boosted task success by 40% and reached Worlds semifinals (top 3%).
- Mentored 5 teams, 2 advancing to World Championship division finals, impacting 50+ students.

FIRST Tech Challenge Dec. 2019 - Feb. 2025
Team Captain Dublin, California

- Managed a total of 22 people across two teams to design, build, and code robots for the FIRST Tech Challenge.
- Award winners at Norcal states 2023 & 2024 (THINK award & CONTROL Award) and top 20 teams in region.
- Presented robot to sponsors such as Northrop Grumman, John Deere, and Automation Anywhere, and raised over \$12,000 in funding

PROJECTS

Guided Vector Fields for Mobile Robot Path Following July 2024 - March 2025
Independent Research

- Researched methods to use centripetal force correction to increase accuracy in Guided Vector Fields for 2D mobile robots.
- Guided vector fields following is a time-independent path following method that is highly resilient to error and capable of supporting real-time obstacle avoidance.

Monte Carlo Localization for Mobile Robots April 2023 - October 2024
FTC Project

- Wrote a Monte Carlo Localization (MCL) library from scratch in Java for use by our team in the FIRST Tech Challenge.
- MCL is a probabilistic localization method that allows combination of multiple sensor readings (LiDAR, odometry, etc) to allow a robot to determine its position with high accuracy

ACHIEVEMENTS

- **1st Place and Most Technical, @ Startup Incubator Hackthon** For our spatially-intelligent dispatch system for first responders, allowing responders to map location in real time. Won \$3000. 2026
- **Winner Advanced Track, Hard Hacks @UCSD** For our multi-layered biometric security system, featuring custom Face ID and Touch ID. Won \$500 2026
- **Best Overall, HelixDB Hackathon** For our use of HelixDB in a knowledge graph generation tool and knowledge management system 2025
- **Second place, Hack Github @SF** For our custom computer use agent and software learning companion 2025

TECHNICAL SKILLS

Software: C, C++, Python, Java, Git,
Robotics: ROS2, Linux (Ubuntu), SLAM, sensor fusion, control theory, LiDAR, computer vision
Embedded Systems: Arduino, Raspberry Pi, ESP32, Teensy, STM32